Ruchir Kalokhe

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Engineer working across backend systems and ML infrastructure: multi-provider GPU inference, model serving, retrieval and ranking, and the billing, security, and reliability layers that keep them in production. Comfortable owning a feature from runtime container to API to evaluation harness.

EXPERIENCE

Gen AI Intern, Backend and ML | In2peta Services Pvt. Ltd. Remote · Dec 2025 – Present

TypeScript, Fastify, PostgreSQL, Prisma, PgBoss, Docker, Python, FastAPI, React, MotherDuck

- Built the GPU platform backend in **TypeScript** and **Fastify** for multi-provider inference across Modal, Koyeb, and RunPod: unified runtime contracts, PgBoss job orchestration on PostgreSQL, provider routing, live endpoint resolution, and failover so long-running jobs keep serving when a provider drops.
- Automated Hugging Face model deployment (Dockerfiles, handlers, build workers, Prisma catalog) and integrated **LTX-2 / LTX-2.3 Distilled** video runtimes with image conditioning and clips up to 60 seconds.
- Implemented usage-based billing and credit reservations across token, character, second, and image modalities (Razorpay, idempotent Prisma workflows that block overspend); hardened Fastify APIs with scoped keys, plan-aware access, webhook verification, and SSRF blocking; patched LiteLLM cached-token pricing.
- Built lead discovery for **GrowthCrew**, In2peta's outreach product: a background worker searches a **~10.6M-row MotherDuck** leads database and live CoreClaw in parallel, dedupes by email and company, and degrades gracefully when either source fails.
- Unified GrowthCrew's Social Studio and lead-gen apps into one **FastAPI** backend and React shell (one JSON store with locking and atomic writes, one error envelope), retiring a duplicate Express server; made AgentMail outreach safe with valid idempotency keys and a reply relay limited to known leads.

PROJECTS

Auscult | Privacy-preserving observability for healthcare AI agents Python, Presidio, spaCy

- Made agent runs auditable without retaining raw PHI: prompts, tool calls, outputs, and errors are sanitized with Presidio/spaCy and deterministic Faker replacements **before** database write.
- Fail-closed capture so sanitizer or queue failures never silently persist unsanitized text; JSON-safe redaction, background writes, replay/export/purge, Alembic migrations, and a PHI eval harness under CI.
- Shipped OpenAI/Anthropic wrappers, LangChain callbacks, and Codex CLI capture, plus a PHI guard that sanitizes files and tool results before agents read them (Claude Code hook); packaged for PyPI.

Candis | LLM candidate discovery and ranking over 100K profiles Python, NumPy, embeddings, evals

- Ranked top candidates from **100K profiles** in **~13 s** on CPU (NumPy path) via honeypot filters, structured fit scoring, availability weighting, cached LLM reranking, and evidence-guided expansion.
- Beat keyword stuffing: **15** injected weak profiles entered the top-100 under keyword matching, **0** under Candis; pairwise LLM audit **83.7% win rate**, hand-qrel NDCG@10 **0.93+**, prompt-injection audit with **0** effect on final ranking.

Hydrous | Bi-temporal agent memory layer (Hack with Hydra) Python, SQLite, HydraDB, MCP

- Built agent memory that keeps time: versioned facts with validity intervals and supersession, so as-of retrieval answers what was true then; runs on SQLite or a local HydraDB OSS graph via OpenCypher.
- Added an extraction quality gate, feedback-loop prevention, evidence-only answers with safe abstention, and a mem0-compatible API with import; exposed via Python, TypeScript SDK, HTTP, and an MCP server.

DealFlow360 | B2B quote-to-cash platform (academic team project) Next.js, Prisma, PostgreSQL

- Built the billing engine on Prisma: billing calendar, proration, idempotent request-key replay, subscriptions, billing runs, and invoices with PDF export; set up the PostgreSQL schema, migrations, and seed.
- Implemented session auth with per-user credentials and role-based access across five roles (route guards, fail-closed screens), plus password reset, admin audit, rate limiting, and session purge.

More: MuseTalk API, TraceLink, SAR Colorization, AssetFlow · github.com/ruxir-ig

TECHNICAL SKILLS

- **Languages** Python, TypeScript, SQL, C
- **Machine Learning** PyTorch, TensorFlow, Hugging Face, Diffusers, CUDA, scikit-learn, Presidio, spaCy
- **Backend** FastAPI, Fastify, Node.js, Next.js, React, REST APIs, MCP servers, PgBoss, asynchronous workers, SDK design
- **Infrastructure** Docker, PostgreSQL, Prisma, SQLite, MotherDuck, Modal, RunPod, Koyeb, GitHub Actions, Linux

EDUCATION

B.E., Artificial Intelligence and Data Science | Savitribai Phule Pune University · PES MCOE Pune, India · Aug 2023 – Jun 2027

- **Open source & research** Contributed 24-bit truecolor to *ascii-view* (C); **maintainer** of *nitch*; GAN SAR-to-RGB colorization in PyTorch (top 25 internal SIH); BB84 QKD simulator.
- **Community** Member of Google Developer Group on Campus (AIML) - organised IMACE 2026 hackathon and events, and served on the judging jury.